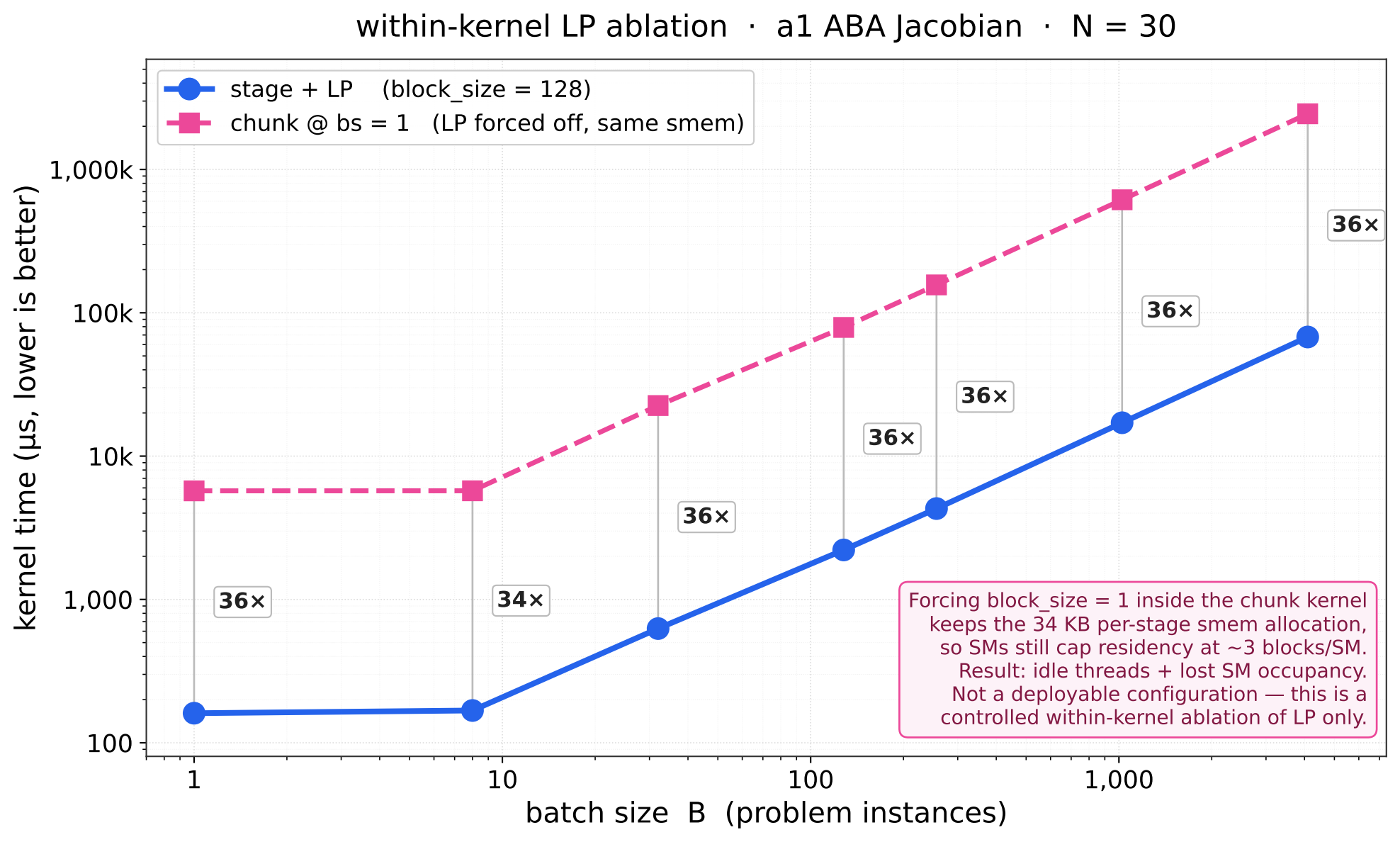
within-kernel LP ablation · a1 ABA Jacobian · N = 30
stage + LP (block_size = 128)
chunk @ bs = 1 (LP forced off, same smem)
kernel time (µs, lower is better)
1,000k
100k
10k
1,000
100
1
10
100
1,000
batch size B (problem instances)
36×
34×
36×
36×
36×
36×
36×
Forcing block_size = 1 inside the chunk kernel keeps the 34 KB per-stage smem allocation, so SMs still cap residency at ~3 blocks/SM. Result: idle threads + lost SM occupancy. Not a deployable configuration — this is a controlled within-kernel ablation of LP only.